

FANUC Robotics Ethernet/IP Connections

Total number of I/O: 4096 can be assigned to Digital I/O Points (*theoretical maximum*).

Most common assignments: 512 and 1024 Digital I/O Points

Ethernet/IP Connections

Number of Ethernet/IP Connections: 64, these are virtual connections to other Ethernet devices.

Motion takes priority

Busy

Step

Hold

Fault

Run

I/O

Prod

TCyc

MYPROG LINE 0

T1

ABORTED

JOINT

100%

I/O EtherNet/IP

EtherNet/IP List(Rack 89)

1/64

Description	TYP	Enable	Status	Slot
Connection1	ADP	FALSE	OFFLINE	1
Connection2	ADP	FALSE	OFFLINE	2
Connection3	ADP	FALSE	OFFLINE	3
Connection4	ADP	FALSE	OFFLINE	4
Connection5	ADP	FALSE	OFFLINE	5
Connection6	ADP	FALSE	OFFLINE	6
Connection7	ADP	FALSE	OFFLINE	7
Connection8	ADP	FALSE	OFFLINE	8
Connection9	ADP	FALSE	OFFLINE	9
ConnectionA	ADP	FALSE	OFFLINE	10

[TYPE]

PING

SAFETY

CONFIG

EDA

To communicate with a PLC the Connection is configured as an **Adapter**.

ADAPTER

An Ethernet/IP Adapter is a device that acts as the endpoint for a network. It provides the inputs and outputs for sensors and actuators, enabling it to exchange real-time data over a network to the PLC controller.

The PLC initiates communication and collects data from one or more adapters. The adapter responds to the PLC's requests and provides the necessary input and output data to control devices like valves, drives, or other machinery.

SCANNER

An EtherNet/IP Scanner is a device that acts as the originator in an EtherNet/IP network, initiating communication sessions with the Adapters to read and write data in real-time.

The scanner controls the flow of information to and from connected devices, configuring connections and managing data transfer at set intervals.

adapter exchanges info with scanner

we need set up plc with robot and robot with plc

FANUC Adapter Configuration

Connection I/O Number: 4032 Digital I/O Points (*theoretical maximum*). *The number of Robot looking at 4 words and each is 16 bits*

Busy Step Hold Fault
Run I/O Prod TCyc MYPROG LINE 0 T1 ABORTED JOINT 100 %

I/O EtherNet/IP

Adapter configuration : 1/3

Description : Connection1
Input size (words) : 4
Output size (words) : 4
Alarm Severity : WARN

Scanner IP : *****
API O=>T : 0
API T=>O : 0

API O and API T - input and output

Only when scanner initiates communication until it get's filled in.

[TYPE] PREV ? HELP

In this example, **4 (16 Bit) Word** are defined for the Digital Inputs and Outputs.

Number of **Digital Inputs:** 64

Number of **Digital Outputs:** 64

Digital I/O Connections cannot exceed to the total configured I/O assignments.

In the PLC's Scanner configuration the robot's Ethernet/IP Module Definition for the example above would be defined as **8 Input Bytes** and **8 Output Bytes**.
(8 bit bytes x 8 = 64)

Module Definition*

Module Connection

EtherNet/IP Standard

Size (8-bit)

Robot Connection(s) Input Output

Standard_Slot_01 8 8

It is using 8 bit bytes
Robot uses 16 bit words

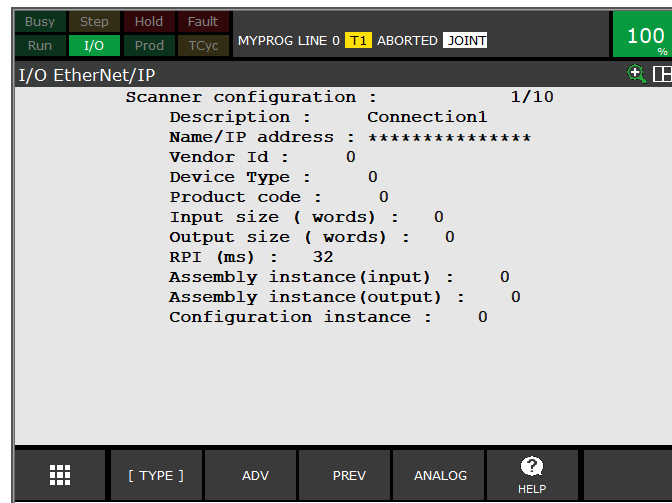
Sizes entered here for standard connections are in bytes (8-bits) while sizes in the robot are entered as words (16-bits).

OK Cancel Help

Scanner Configuration

The robot does have the capabilities of being configured as a Scanner, this is usually done for non-PLC devices.

The Scanner requires the configuration information to initiate communications with an adapter. See the FANUC Ethernet/IP Setup and Operations Manual.



Robot can be set up as a scanner.